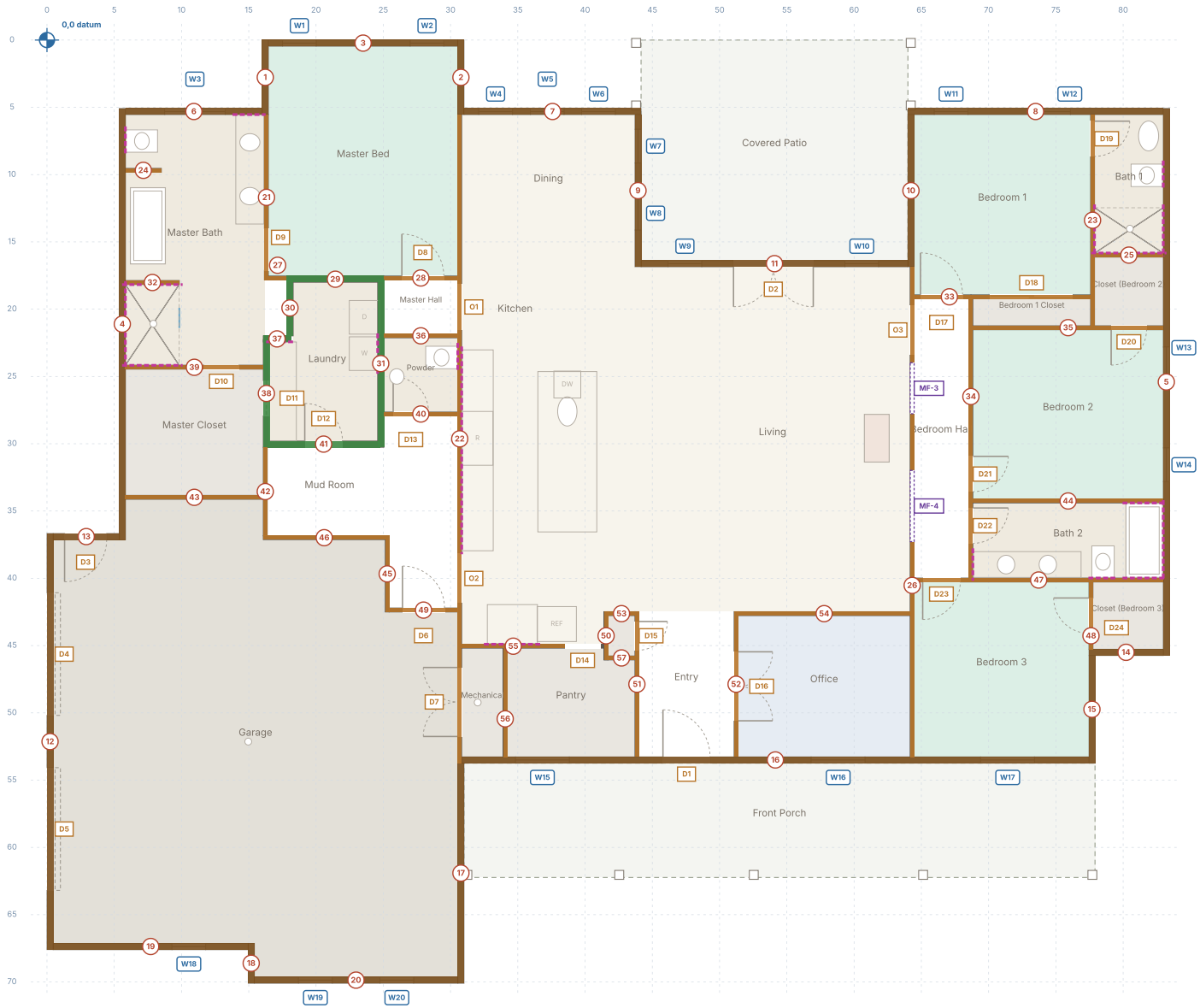
REAR · PATIO



FRONT · PORCH

2×6 exterior @ 16" · 2×4 partition @ 16" · 2×6 plate, staggered 2×4 studs · blocking band · wall # · D# door, rough opening · W# window · MF- recessed manifold cabinet (R-1) · 0,0 datum, grid 5'

Exterior 2×6 (brown), partitions 2×4 (tan), laundry sound walls 2×6 plate with staggered 2×4 studs (green). · Tags: wall numbers in red circles, doors in tan boxes, windows in blue. · Roof trusses bear on the exterior walls and girder G1; every interior wall is non-bearing.

draft

S-1 Framing · 64 Stone Sheep Circle

Wall types, rough openings, headers, blocking and the roof truss layout. Walls numbered from 1, exterior first. App version v28. From the owner's plans (Larson House, 6/11/26).

Notes

1. Datum 0,0 (blue target, top left of the drawing): where the outside face of the garage-door wall (X = 0) meets the line of the outside face of the master bedroom's rear wall (Y = 0). It is the rear-left corner of the layout string lines, not a wall corner: set it from those two lines. The grid is 5'-0" each way from it; X runs to the right (toward the bedroom wing), Y toward the front (the porch). Every sheet uses it.
2. Walls: 2×6 exterior, 2×4 interior (owner, 2026-10-13), drawn at stud depth (5-1/2" and 3-1/2"). Plumbing walls are 2×4 like the rest; where a 3" drain or vent goes through a plate, strap it (IRC R602.6.1).
3. Laundry sound walls (green, owner 2026-10-13): 2×6 plates with 2×4 studs alternating faces at 12" o.c. (24" o.c. on each face), R-15 batt woven between, no stud touching both faces. They grow 2" into the laundry, so the master bedroom, master closet, mud room, master hall and powder room keep their A-1 sizes; the laundry is 2" smaller each way. D11 needs a 2×6 pocket frame.
4. Plate height 9'-1 1/8" (PT plate, 104-5/8" precut studs, double top plate) for 9'-0" ceilings. The kitchen and living vault is in the trusses (4:12 scissor bottom chords); no wall runs past the plate.
5. Headers: with 8' heads there is 7-5/8" between the rough opening and the top plates, so exterior headers are 7-1/4" deep: 2-2×8 up to a 4' opening, LVL above that (the two 9' garage doors and the 8' patio French doors: engineer to confirm, or raise the plate). Interior walls carry no roof: a flat 2×4 header with cripples (IRC R602.7.4).
6. Roof: trusses at 24" o.c. bearing on the exterior walls (roof framing schedule). The main trusses span front to rear on the A-2 ridge line; a 2-ply girder truss (G1) on the master-wing/kitchen line carries the master wing and garage trusses, so the wall under it is a partition. Hang every partition top plate with slip clips (no nails into the bottom chords) so truss uplift doesn't crack the ceilings.
7. Door rough openings: leaf + 2" wide, + 2-1/2" tall; pockets 2 x leaf + 1". Window rough openings: unit + 1/2" each way (confirm with the window maker).

Open items

- Manifold cabinet depth: a 2×4 wall gives 3-1/2". Recessed manifold cabinets usually need 3-3/4" to 4-3/4"; pick the manifold and cabinet first. If it needs more, fur that stretch of wall 26 to 2×6 (it bumps 2" into the living room).
- MF-3: E-1 R41 (bedroom hall, 24") is in the leader bay under the cabinet; E-1 R47 (living, 16") is in the leader bay under the cabinet; E-1 S16 (living, 48") is inside the rough opening or its header (clear at 53" and up); E-1 S19 (bedroom hall, 48") is inside the rough opening or its header (clear at 53" and up); the fireplace insert is 1'-6 1/2" off the other face: keep the cabinet and its tubing out of the insert's clearances and chase (its manual).
- MF-4: E-1 R48 (living, 16") is in the leader bay under the cabinet; E-1 R51 (living, 8") is in the leader bay under the cabinet; the fireplace insert is 1'-6 1/2" off the other face: keep the cabinet and its tubing out of the insert's clearances and chase (its manual).
- Datum: confirm the 0,0 layout corner with the surveyor's pins before the forms go in.
- Window sizes and heights are scaled from the elevations; confirm the window schedule with the supplier before framing.
- Bedroom 3's only window (W17, 4'x4') scales with a 4'-0" sill: above the 44" egress limit. Lower it or use a casement sized for egress.
- Interior door heights are not on the plans (8'-0" assumed to match the exterior doors on the elevations).
- Roof trusses: the layout here is a draft from A-2. The truss maker designs the trusses, the hip and valley sets, the G1 girder and its end posts, and the over-framed 8:12 gables at the entry and office.
- Heel: 12" energy heel assumed (full R-49 to the wall line); it sets the eave height on the elevations.
- Fireplace insert: chimney chase framing through the vaulted living ceiling and roof; clearances per the insert's manual.
- Living room label on A-1 says 17'-4" wide; the walls measure about 20'-5". Framing follows the walls.

8. Garage separation: 5/8" Type X on the garage side of the walls to the house and on the garage ceiling; D6 is a 20-minute self-closing door.
9. Recessed manifold cabinets (purple, R-1): MF-3 and MF-4 sit in 2×4 partitions. Each is a framed opening: kings full height, a flat 2×4 header with cripples above, a flat 2×4 sill, and no cripples below it because the loop leaders come up out of the slab in that bay. Bore the bottom plate and sill 7/8" at each port; nothing else goes in the bay (no boxes, cable or pipe).
10. Blocking (magenta): the bands on the drawing and the blocking schedule; grab-bar blocking at every toilet, tub and shower; vanity tops and mirrors, the washer box, the TV over the fireplace.

- Foundation (slab assumed: the garage and mechanical floor drains, the radiant slab), thickened footings under G1's end posts and the patio and porch posts, anchor schedule.

Wall schedule

Wall	Type	Length	Between	Openings	Studs	Notes
1	2×6 exterior	5'-6 1/2"	outside, Master Bed	—	10	PT bottom plate on the slab, sill gasket, 1/2" anchor bolts at 6' o.c. and 12" from ends
2	2×6 exterior	5'-6 1/2"	Master Bed, outside	—	10	PT bottom plate on the slab, sill gasket, 1/2" anchor bolts at 6' o.c. and 12" from ends
3	2×6 exterior	14'-1"	outside, Master Bed	W1, W2	24	PT bottom plate on the slab, sill gasket, 1/2" anchor bolts at 6' o.c. and 12" from ends
4	2×6 exterior	32'-1/2"	outside, Master Bath, Master Closet	—	30	PT bottom plate on the slab, sill gasket, 1/2" anchor bolts at 6' o.c. and 12" from ends; blocking: grab bars and paper holder (ADA toilet), grab bars, valve and bench (Shower 4' x 6' (glass door))
5	2×6 exterior	40'-8"	Bath 1, outside, Bedroom 2, Bath 2	W13, W14	44	PT bottom plate on the slab, sill gasket, 1/2" anchor bolts at 6' o.c. and 12" from ends; blocking: grab bars and paper holder (ADA toilet), grab bars, valve and bench (Shower 5' x 3'-4"), grab bars and valve (Standard tub 1 [66W])
6	2×6 exterior	10'-2"	outside, Master Bath	W3	17	PT bottom plate on the slab, sill gasket, 1/2" anchor bolts at 6' o.c. and 12" from ends; blocking: cabinets and vanity top (Double vanity, 8')
7	2×6 exterior	13'-2"	outside, Dining	W4, W5, W6	27	PT bottom plate on the slab, sill gasket, 1/2" anchor bolts at 6' o.c. and 12" from ends
8	2×6 exterior	19'-0"	outside, Bedroom 1, Bath 1	W11, W12	28	PT bottom plate on the slab, sill gasket, 1/2" anchor bolts at 6' o.c. and 12" from ends
9	2×6 exterior	11'-3 1/2"	Dining, Covered Patio	W7, W8	22	PT bottom plate on the slab, sill gasket, 1/2" anchor bolts at 6' o.c. and 12" from ends
10	2×6 exterior	11'-3 1/2"	Covered Patio, Bedroom 1	—	14	PT bottom plate on the slab, sill gasket, 1/2" anchor bolts at 6' o.c. and 12" from ends
11	2×6 exterior	19'-10"	Covered Patio, Living	W9, W10, D2	31	PT bottom plate on the slab, sill gasket, 1/2" anchor bolts at 6' o.c. and 12" from ends
12	2×6 exterior	30'-11"	outside, Garage	D4, D5	31	PT bottom plate on the slab, sill gasket, 1/2" anchor bolts at 6' o.c. and 12" from ends
13	2×6 exterior	4'-11"	outside, Garage	D3	12	PT bottom plate on the slab, sill gasket, 1/2" anchor bolts at 6' o.c. and 12" from ends
14	2×6 exterior	5'-6"	Closet (Bedroom 3), outside	—	10	PT bottom plate on the slab, sill gasket, 1/2" anchor bolts at 6' o.c. and 12" from ends

Wall	Type	Length	Between	Openings	Studs	Notes
15	2×6 exterior	8'-0"	Bedroom 3, outside	—	11	PT bottom plate on the slab, sill gasket, 1/2" anchor bolts at 6' o.c. and 12" from ends
16	2×6 exterior	46'-7 1/2"	Pantry, Front Porch, Office, Bedroom 3	W15, W16, W17, D1	55	PT bottom plate on the slab, sill gasket, 1/2" anchor bolts at 6' o.c. and 12" from ends
17	2×6 exterior	16'-4 1/2"	Garage, Front Porch, outside	—	18	PT bottom plate on the slab, sill gasket, 1/2" anchor bolts at 6' o.c. and 12" from ends
18	2×6 exterior	2'-11 1/2"	outside, Garage	—	8	PT bottom plate on the slab, sill gasket, 1/2" anchor bolts at 6' o.c. and 12" from ends
19	2×6 exterior	14'-6"	Garage, outside	W18	20	PT bottom plate on the slab, sill gasket, 1/2" anchor bolts at 6' o.c. and 12" from ends
20	2×6 exterior	15'-1 1/2"	Garage, outside	W19, W20	25	PT bottom plate on the slab, sill gasket, 1/2" anchor bolts at 6' o.c. and 12" from ends
21	2×4 partition	12'-3 1/2"	Master Bath, Master Bed	D9	18	plumbing wall, 2×4 (owner): 3" drains and vents bored through plates get a steel strap across the cut (IRC R602.6.1)
22	2×4 partition	48'-2 1/2"	Kitchen, Mud Room	D7, O1, O2	50	plumbing wall, 2×4 (owner): 3" drains and vents bored through plates get a steel strap across the cut (IRC R602.6.1); blocking: grab bars and paper holder (ADA toilet), cabinets and vanity top (Wall cabinets and counter, left wall)
23	2×4 partition	15'-8 1/2"	Bedroom 1, Bath 1, Closet (Bedroom 2)	D19	20	plumbing wall, 2×4 (owner): 3" drains and vents bored through plates get a steel strap across the cut (IRC R602.6.1); blocking: grab bars, valve and bench (Shower 5' x 3'-4")
24	2×4 partition	2'-8"	Master Bath	—	8	—
25	2×4 partition	5'-1 1/2"	Bath 1, Closet (Bedroom 2)	—	9	plumbing wall, 2×4 (owner): 3" drains and vents bored through plates get a steel strap across the cut (IRC R602.6.1); blocking: grab bars, valve and bench (Shower 5' x 3'-4")
26	2×4 partition	36'-5"	Living, Bedroom Hall, Office, Bedroom 3	O3, MF-3, MF-4	37	recessed manifold cabinet MF-3, MF-4 (R-1): framed opening, see the recess schedule
27	2×4 partition	1'-4 1/2"	Master Bed	—	7	—
28	2×4 partition	5'-6"	Master Bed, Master Hall	D8	13	—
29	2×6 plate, staggered 2×4 studs	7'-2 1/2"	Master Bed, Laundry	—	13	sound wall: 2×6 plates, 2×4 studs alternating faces at 12" o.c. (24" o.c. each face), R-15 batt woven between, no stud touches both faces; grown 2" into the laundry
30	2×6 plate, staggered 2×4 studs	4'-2"	Laundry	—	10	sound wall: 2×6 plates, 2×4 studs alternating faces at 12" o.c. (24" o.c. each face), R-15 batt woven between, no stud touches both faces; grown 2" into the laundry
31	2×6 plate, staggered 2×4 studs	12'-5"	Laundry, Master Hall, Powder	—	18	sound wall: 2×6 plates, 2×4 studs alternating faces at 12" o.c. (24" o.c. each face), R-15 batt woven between, no stud touches both faces; grown 2" into the laundry; blocking: washer box (Washer (curved front loading))
32	2×4 partition	4'-0"	Master Bath	—	8	plumbing wall, 2×4 (owner): 3" drains and vents bored through plates get a steel strap across the cut (IRC R602.6.1); blocking: grab bars, valve and bench (Shower 4' x 6' (glass door))
33	2×4 partition	13'-1 1/2"	Bedroom 1, Bedroom Hall, Bedroom 1 Closet	D17, D18	19	—

Wall	Type	Length	Between	Openings	Studs	Notes
34	2×4 partition	20'-9"	Bedroom Hall, Bedroom 2, Bath 2	D21, D22	27	blocking: cabinets and vanity top (Double vanity, 8')
35	2×4 partition	14'-2"	Bedroom 1 Closet, Bedroom 2, Closet (Bedroom 2)	D20	19	—
36	2×4 partition	5'-6"	Master Hall, Powder	—	10	plumbing wall, 2×4 (owner): 3" drains and vents bored through plates get a steel strap across the cut (IRC R602.6.1)
37	2×6 plate, staggered 2×4 studs	2'-1/2"	Laundry	—	8	sound wall: 2×6 plates, 2×4 studs alternating faces at 12" o.c. (24" o.c. each face), R-15 batt woven between, no stud touches both faces; grown 2" into the laundry; blocking: cabinets and vanity top (Laundry cabinets / counter)
38	2×6 plate, staggered 2×4 studs	8'-0"	Laundry, Master Closet	D11	13	sound wall: 2×6 plates, 2×4 studs alternating faces at 12" o.c. (24" o.c. each face), R-15 batt woven between, no stud touches both faces; grown 2" into the laundry; the pocket door D11 needs a 2×6 pocket frame here
39	2×4 partition	10'-3"	Master Bath, Master Closet	D10	17	plumbing wall, 2×4 (owner): 3" drains and vents bored through plates get a steel strap across the cut (IRC R602.6.1); blocking: grab bars, valve and bench (Shower 4' x 6' (glass door))
40	2×4 partition	5'-6"	Powder, Mud Room	D13	13	—
41	2×6 plate, staggered 2×4 studs	8'-4 1/2"	Laundry, Mud Room	D12	17	sound wall: 2×6 plates, 2×4 studs alternating faces at 12" o.c. (24" o.c. each face), R-15 batt woven between, no stud touches both faces; grown 2" into the laundry
42	2×4 partition	6'-7"	Master Closet, Mud Room, Garage	—	10	garage separation: 5/8" Type X on the garage side, seal penetrations
43	2×4 partition	10'-3"	Master Closet, Garage	—	13	garage separation: 5/8" Type X on the garage side, seal penetrations
44	2×4 partition	14'-2"	Bedroom 2, Bath 2	—	16	plumbing wall, 2×4 (owner): 3" drains and vents bored through plates get a steel strap across the cut (IRC R602.6.1); blocking: grab bars and valve (Standard tub 1 [66W])
45	2×4 partition	5'-8"	Garage, Mud Room	—	10	garage separation: 5/8" Type X on the garage side, seal penetrations
46	2×4 partition	9'-1"	Mud Room, Garage	—	12	garage separation: 5/8" Type X on the garage side, seal penetrations
47	2×4 partition	18'-6"	Bath 2, Bedroom 3, Closet (Bedroom 3)	D23	22	plumbing wall, 2×4 (owner): 3" drains and vents bored through plates get a steel strap across the cut (IRC R602.6.1); blocking: grab bars and paper holder (ADA toilet), grab bars and valve (Standard tub 1 [66W])
48	2×4 partition	5'-0"	Bedroom 3, Closet (Bedroom 3)	D24	12	—
49	2×4 partition	5'-1"	Mud Room, Garage	D6	12	garage separation: 5/8" Type X on the garage side, seal penetrations
50	2×4 partition	3'-7"	Kitchen, Coat Closet	—	8	—
51	2×4 partition	10'-9 1/2"	Coat Closet, Entry, Pantry	D15	18	—
52	2×4 partition	10'-9 1/2"	Entry, Office	D16	16	—

Wall	Type	Length	Between	Openings	Studs	Notes
53	2×4 partition	2'-0"	Kitchen	—	7	—
54	2×4 partition	12'-9 1/2"	Living, Office	—	15	—
55	2×4 partition	7'-8"	Kitchen, Mechanical, Pantry	—	11	blocking: cabinets and vanity top (Counter, front wall)
56	2×4 partition	8'-1"	Mechanical, Pantry	—	12	—
57	2×4 partition	2'-0"	Pantry	—	7	—

Door schedule

Door	Name	Type	Leaf	Rough opening	Header	Between
D1	Front entry	exterior entry door	3'-0" x 8'-0"	3'-2" x 8'-2 1/2"	2-2×8 + 1/2" ply, 1 jack each end	Entry / Front Porch
D2	Patio French doors	French door pair	6'-0" x 8'-0"	6'-2" x 8'-2 1/2"	3-1/2" x 7-1/4" LVL (2-ply 1-3/4"), 2 jacks each end	Living / Covered Patio
D3	Garage service door	exterior door	3'-0" x 6'-8"	3'-2" x 6'-10 1/2"	2-2×8 + 1/2" ply, 1 jack each end	Garage / outside
D4	Garage door 1	overhead garage door	9'-0" x 8'-0"	9'-0" x 8'-0" (door maker's RO)	5-1/4" x 7-1/4" LVL (3-ply), 2 jacks each end; engineer to confirm	Garage / outside
D5	Garage door 2	overhead garage door	9'-0" x 8'-0"	9'-0" x 8'-0" (door maker's RO)	5-1/4" x 7-1/4" LVL (3-ply), 2 jacks each end; engineer to confirm	Garage / outside
D6	Garage to mud room	20-min fire-rated, self-closing	3'-0" x 8'-0"	3'-2" x 8'-2 1/2"	flat 2×4, cripples at 16" (non-bearing)	Garage / Mud Room
D7	Mechanical (from garage)	door pair	5'-0" x 8'-0"	5'-2" x 8'-2 1/2"	flat 2×4, cripples at 16" (non-bearing)	Garage / Mechanical
D8	Master bedroom	swing door	3'-0" x 8'-0"	3'-2" x 8'-2 1/2"	flat 2×4, cripples at 16" (non-bearing)	Master Hall / Master Bed
D9	Master bed to bath	cased opening	3'-0" x 8'-0"	3'-0" x 8'-0" cased	flat 2×4, cripples at 16" (non-bearing)	Master Bed / Master Bath
D10	Master bath to closet	cased opening	2'-6" x 8'-0"	2'-6" x 8'-0" cased	flat 2×4, cripples at 16" (non-bearing)	Master Bath / Master Closet
D11	Master closet to laundry	pocket door	2'-6" x 8'-0"	5'-1" x 8'-2 1/2" (pocket frame)	flat 2×4, cripples at 16" (non-bearing)	Master Closet / Laundry
D12	Laundry	swing door	2'-8" x 8'-0"	2'-10" x 8'-2 1/2"	flat 2×4, cripples at 16" (non-bearing)	Mud Room / Laundry
D13	Powder	swing door	2'-6" x 8'-0"	2'-8" x 8'-2 1/2"	flat 2×4, cripples at 16" (non-bearing)	Mud Room / Powder

Door	Name	Type	Leaf	Rough opening	Header	Between
D14	Pantry	cased opening	2'-6" x 8'-0"	2'-6" x 8'-0" cased	flat 2×4, cripples at 16" (non-bearing)	Kitchen / Pantry
D15	Coat closet	swing door	2'-0" x 8'-0"	2'-2" x 8'-2 1/2"	flat 2×4, cripples at 16" (non-bearing)	Entry / Coat Closet
D16	Office French doors	French door pair	5'-0" x 8'-0"	5'-2" x 8'-2 1/2"	flat 2×4, cripples at 16" (non-bearing)	Entry / Office
D17	Bedroom 1	swing door	3'-0" x 8'-0"	3'-2" x 8'-2 1/2"	flat 2×4, cripples at 16" (non-bearing)	Bedroom Hall / Bedroom 1
D18	Bedroom 1 closet	bypass (sliding) pair	6'-0" x 8'-0"	6'-2" x 8'-2 1/2"	flat 2×4, cripples at 16" (non-bearing)	Bedroom 1 / Bedroom 1 Closet
D19	Bath 1	swing door	2'-6" x 8'-0"	2'-8" x 8'-2 1/2"	flat 2×4, cripples at 16" (non-bearing)	Bedroom 1 / Bath 1
D20	Bedroom 2 closet	swing door	2'-6" x 8'-0"	2'-8" x 8'-2 1/2"	flat 2×4, cripples at 16" (non-bearing)	Bedroom 2 / Closet (Bedroom 2)
D21	Bedroom 2	swing door	2'-6" x 8'-0"	2'-8" x 8'-2 1/2"	flat 2×4, cripples at 16" (non-bearing)	Bedroom Hall / Bedroom 2
D22	Bath 2	swing door	2'-6" x 8'-0"	2'-8" x 8'-2 1/2"	flat 2×4, cripples at 16" (non-bearing)	Bedroom Hall / Bath 2
D23	Bedroom 3	swing door	2'-8" x 8'-0"	2'-10" x 8'-2 1/2"	flat 2×4, cripples at 16" (non-bearing)	Bedroom Hall / Bedroom 3
D24	Bedroom 3 closet	swing door	2'-6" x 8'-0"	2'-8" x 8'-2 1/2"	flat 2×4, cripples at 16" (non-bearing)	Bedroom 3 / Closet (Bedroom 3)
O1	Kitchen to master hall	cased opening	3'-5" x 8'-0"	3'-5" x 8'-0" cased	flat 2×4, cripples at 16" (non-bearing)	Kitchen / Master Hall
O2	Kitchen to mud room	cased opening	3'-7" x 8'-0"	3'-7" x 8'-0" cased	flat 2×4, cripples at 16" (non-bearing)	Kitchen / Mud Room
O3	Living to bedroom hall	cased opening	3'-7" x 8'-0"	3'-7" x 8'-0" cased	flat 2×4, cripples at 16" (non-bearing)	Living / Bedroom Hall

Window schedule

Window	Room	Unit	Rough opening	Head / sill	Header	Egress
W1	Master Bed	2'-6" x 5'-0"	2'-6 1/2" x 5'-1/2"	8'-0" / 3'-0"	2-2×8 + 1/2" ply, 1 jack each end	Bedroom: needs 5.7 sf net opening, 20" clear width, 24" clear height, sill ≤ 44" (sill here 3'-0")
W2	Master Bed	2'-6" x 5'-0"	2'-6 1/2" x 5'-1/2"	8'-0" / 3'-0"	2-2×8 + 1/2" ply, 1 jack each end	Bedroom: needs 5.7 sf net opening, 20" clear width, 24" clear height, sill ≤ 44" (sill here 3'-0")
W3	Master Bath	4'-6" x 3'-0"	4'-6 1/2" x 3'-1/2"	8'-0" / 5'-0"	3-1/2" x 7-1/4" LVL (2-ply 1-3/4"), 2 jacks each end	—

Window	Room	Unit	Rough opening	Head / sill	Header	Egress
W4	Dining	2'-6" x 5'-0"	2'-6 1/2" x 5'-1/2"	8'-0" / 3'-0"	2-2×8 + 1/2" ply, 1 jack each end	—
W5	Dining	2'-6" x 5'-0"	2'-6 1/2" x 5'-1/2"	8'-0" / 3'-0"	2-2×8 + 1/2" ply, 1 jack each end	—
W6	Dining	2'-6" x 5'-0"	2'-6 1/2" x 5'-1/2"	8'-0" / 3'-0"	2-2×8 + 1/2" ply, 1 jack each end	—
W7	Dining	2'-6" x 5'-0"	2'-6 1/2" x 5'-1/2"	8'-0" / 3'-0"	2-2×8 + 1/2" ply, 1 jack each end	—
W8	Dining	2'-6" x 5'-0"	2'-6 1/2" x 5'-1/2"	8'-0" / 3'-0"	2-2×8 + 1/2" ply, 1 jack each end	—
W9	Living	2'-6" x 5'-0"	2'-6 1/2" x 5'-1/2"	8'-0" / 3'-0"	2-2×8 + 1/2" ply, 1 jack each end	—
W10	Living	2'-6" x 5'-0"	2'-6 1/2" x 5'-1/2"	8'-0" / 3'-0"	2-2×8 + 1/2" ply, 1 jack each end	—
W11	Bedroom 1	2'-6" x 5'-0"	2'-6 1/2" x 5'-1/2"	8'-0" / 3'-0"	2-2×8 + 1/2" ply, 1 jack each end	Bedroom: needs 5.7 sf net opening, 20" clear width, 24" clear height, sill ≤ 44" (sill here 3'-0")
W12	Bedroom 1	2'-6" x 5'-0"	2'-6 1/2" x 5'-1/2"	8'-0" / 3'-0"	2-2×8 + 1/2" ply, 1 jack each end	Bedroom: needs 5.7 sf net opening, 20" clear width, 24" clear height, sill ≤ 44" (sill here 3'-0")
W13	Bedroom 2	2'-6" x 5'-0"	2'-6 1/2" x 5'-1/2"	8'-0" / 3'-0"	2-2×8 + 1/2" ply, 1 jack each end	Bedroom: needs 5.7 sf net opening, 20" clear width, 24" clear height, sill ≤ 44" (sill here 3'-0")
W14	Bedroom 2	2'-6" x 5'-0"	2'-6 1/2" x 5'-1/2"	8'-0" / 3'-0"	2-2×8 + 1/2" ply, 1 jack each end	Bedroom: needs 5.7 sf net opening, 20" clear width, 24" clear height, sill ≤ 44" (sill here 3'-0")
W15	Pantry	4'-0" x 4'-0"	4'-1/2" x 4'-1/2"	8'-0" / 4'-0"	3-1/2" x 7-1/4" LVL (2-ply 1-3/4"), 2 jacks each end	—
W16	Office	4'-0" x 4'-0"	4'-1/2" x 4'-1/2"	8'-0" / 4'-0"	3-1/2" x 7-1/4" LVL (2-ply 1-3/4"), 2 jacks each end	—
W17	Bedroom 3	4'-0" x 4'-0"	4'-1/2" x 4'-1/2"	8'-0" / 4'-0"	3-1/2" x 7-1/4" LVL (2-ply 1-3/4"), 2 jacks each end	Bedroom: needs 5.7 sf net opening, 20" clear width, 24" clear height, sill ≤ 44" (sill here 4'-0" — too high)
W18	Garage	2'-6" x 5'-0"	2'-6 1/2" x 5'-1/2"	8'-0" / 3'-0"	2-2×8 + 1/2" ply, 1 jack each end	—
W19	Garage	2'-6" x 5'-0"	2'-6 1/2" x 5'-1/2"	8'-0" / 3'-0"	2-2×8 + 1/2" ply, 1 jack each end	—
W20	Garage	2'-6" x 5'-0"	2'-6 1/2" x 5'-1/2"	8'-0" / 3'-0"	2-2×8 + 1/2" ply, 1 jack each end	—

Roof framing schedule

Draft truss layout from A-2 (see the 3D model's roof trusses layer). Stations are feet from the datum. The truss maker designs the final layout.

Mark	Roof	Type	Span	Pitch	Peak above slab	Count	Where
G1	Main roof (great room, bedroom wing)	2-ply girder, carries the master wing and garage trusses	48'-8"	6:12	22'-2"	1	X 30.8 to 30.8
A1	Main roof (great room, bedroom wing)	scissor, 4:12 bottom chord (vault)	48'-8"	6:12 / 4:12	22'-2"	6	X 32.8 to 42.8
A2	Main roof (great room, bedroom wing)	scissor, 4:12 bottom chord (vault), tall end	37'-4"	6:12 / 4:12	22'-2"	8	X 44.8 to 58.8
A3	Main roof (great room, bedroom wing)	scissor, 4:12 bottom chord (vault), tall end	37'-4"	6:12 / 4:12	21'-4"	1	X 60.8 to 60.8
A4	Main roof (great room, bedroom wing)	scissor, 4:12 bottom chord (vault), tall end	37'-4"	6:12 / 4:12	20'-4"	1	X 62.8 to 62.8
A5	Main roof (great room, bedroom wing)	hip set, stepped flat tops	48'-8"	6:12	13'-4" to 19'-4"	7	X 64.8 to 76.8
A6	Main roof (great room, bedroom wing)	hip set, stepped flat tops, tall end	40'-8"	6:12	10'-4" to 12'-4"	3	X 78.8 to 82.8
A7	Main roof (great room, bedroom wing)	hip set, stepped flat tops	40'-8"	6:12	10'-1/2"	1	X 83.4 to 83.4
B1	Master wing	hip set, stepped flat tops	25'-4 1/2"	6:12	10'-1/2" to 16'-1/2"	7	Y 5.1 to 17.1
B2	Master wing	common	25'-4 1/2"	6:12	16'-4"	9	Y 19.1 to 34.1
C1	Master bedroom bump-out	hip set, stepped flat tops	15'-0"	6:12	10'-1/2" to 12'-6"	4	Y 0.1 to 5.0
D1	Garage	hip set, stepped flat tops	25'-4 1/2"	6:12	10'-1/2"	1	Y 34.2 to 34.2
D2	Garage	hip set, stepped flat tops, tall end	25'-4 1/2"	6:12	11'-1/2"	1	Y 36.2 to 36.2
D3	Garage	hip set, stepped flat tops	30'-9"	6:12	11'-11 1/2" to 17'-1/2"	12	Y 38.2 to 66.2
D4	Garage	common	30'-9"	6:12	17'-8"	3	Y 50.2 to 54.2
D5	Garage	hip set, stepped flat tops, tall end	15'-9 1/2"	6:12	10'-11 1/2"	1	Y 68.2 to 68.2
D6	Garage	hip set, stepped flat tops	15'-9 1/2"	6:12	10'-1/2"	1	Y 70.0 to 70.0
PR	Patio	2×10 rafters on ridge and side beams	20'-5"	6:12	14'-1 1/2"	20	Y 0.2 to 16.3, a pair per station
PO	Front porch	2×8 rafters, ledger to beam	8'-3 1/2"	2:12	10'-4 1/2"	25	X 31.1 to 77.9

Recessed manifold cabinets

R-1's manifold cabinets that sit in the 2×4 partitions. MF-1, MF-2 and MF-G are surface-mounted and need no framing.

Cabinet	Room	Wall	Cabinet	Rough opening	Framing	Conflicts
MF-3	Bedroom Hall	wall 26	3'-9" wide x 1'-7" tall, bottom at 2'-6" (R-1)	3'-9 1/2" wide x 1'-7 1/2" tall, 2'-5 1/2" to 4'-1 1/2" above the slab	kings, flat 2×4 header and sill, open leader bay	E-1 R41 (bedroom hall, 24") is in the leader bay under the cabinet; E-1 R47 (living, 16") is in the leader bay under the cabinet; E-1 S16 (living, 48") is inside the rough opening or its header (clear at 53" and up); E-1 S19 (bedroom hall, 48") is inside the rough opening or its header (clear at 53" and up); the fireplace insert is 1'-6 1/2" off the other face: keep the cabinet and its tubing out of the insert's clearances and chase (its manual)
MF-4	Bedroom Hall	wall 26	5'-3" wide x 1'-7" tall, bottom at 2'-6" (R-1)	5'-3 1/2" wide x 1'-7 1/2" tall, 2'-5 1/2" to 4'-1 1/2" above the slab	kings, flat 2×4 header and sill, open leader bay	E-1 R48 (living, 16") is in the leader bay under the cabinet; E-1 R51 (living, 8") is in the leader bay under the cabinet; the fireplace insert is 1'-6 1/2" off the other face: keep the cabinet and its tubing out of the insert's clearances and chase (its manual)

Blocking schedule

Wall	Room	Fixture	For	Blocking
wall 4	Master Bath	ADA toilet	Grab bars and paper holder	2×8 flat between studs, 33"-36" AFF, plus 2×6 at 24" for the paper holder
wall 4	Master Bath	Shower 4' x 6' (glass door)	Grab bars, valve and bench	2×8 at 33"-36" and 2×6 at 48" AFF on every wet wall; valve blocking per the valve
wall 5	Bath 1	ADA toilet	Grab bars and paper holder	2×8 flat between studs, 33"-36" AFF, plus 2×6 at 24" for the paper holder
wall 5	Bath 1	Shower 5' x 3'-4"	Grab bars, valve and bench	2×8 at 33"-36" and 2×6 at 48" AFF on every wet wall; valve blocking per the valve
wall 5	Bath 2	Standard tub 1 [66W]	Grab bars and valve	2×8 at 33"-36" AFF along the tub and at the valve end
wall 6	Master Bath	Double vanity, 8'	Cabinets and vanity top	2×6 at 34" AFF for the top or counter, 2×6 at 84" for wall cabinets or a mirror
wall 22	Powder	ADA toilet	Grab bars and paper holder	2×8 flat between studs, 33"-36" AFF, plus 2×6 at 24" for the paper holder
wall 22	Kitchen	Wall cabinets and counter, left wall	Cabinets and vanity top	2×6 at 34" AFF for the top or counter, 2×6 at 84" for wall cabinets or a mirror
wall 23	Bath 1	Shower 5' x 3'-4"	Grab bars, valve and bench	2×8 at 33"-36" and 2×6 at 48" AFF on every wet wall; valve blocking per the valve
wall 25	Bath 1	Shower 5' x 3'-4"	Grab bars, valve and bench	2×8 at 33"-36" and 2×6 at 48" AFF on every wet wall; valve blocking per the valve
wall 31	Laundry	Washer (curved front loading)	Washer box	2×6 at 36"-48" AFF for the recessed washer box
wall 32	Master Bath	Shower 4' x 6' (glass door)	Grab bars, valve and bench	2×8 at 33"-36" and 2×6 at 48" AFF on every wet wall; valve blocking per the valve
wall 34	Bath 2	Double vanity, 8'	Cabinets and vanity top	2×6 at 34" AFF for the top or counter, 2×6 at 84" for wall cabinets or a mirror
wall 37	Laundry	Laundry cabinets / counter	Cabinets and vanity top	2×6 at 34" AFF for the top or counter, 2×6 at 84" for wall cabinets or a mirror
wall 39	Master Bath	Shower 4' x 6' (glass door)	Grab bars, valve and bench	2×8 at 33"-36" and 2×6 at 48" AFF on every wet wall; valve blocking per the valve
wall 44	Bath 2	Standard tub 1 [66W]	Grab bars and valve	2×8 at 33"-36" AFF along the tub and at the valve end
wall 47	Bath 2	ADA toilet	Grab bars and paper holder	2×8 flat between studs, 33"-36" AFF, plus 2×6 at 24" for the paper holder
wall 47	Bath 2	Standard tub 1 [66W]	Grab bars and valve	2×8 at 33"-36" AFF along the tub and at the valve end
wall 55	Kitchen	Counter, front wall	Cabinets and vanity top	2×6 at 34" AFF for the top or counter, 2×6 at 84" for wall cabinets or a mirror

Lumber, sheathing and drywall

Rough-order quantities from the wall data, for pricing. The framer takes off the final list.

Item	Quantity	Basis	Notes
Exterior wall, 2×6	328 lf	447 studs 2×6 104-5/8" precut (9'-1 1/8" plate)	PT 2×6 bottom plate + double top plate
Partitions, 2×4	333 lf	476 studs 2×4 precut	PT bottom plate on slab
Laundry sound walls	42 lf	79 studs 2×4 precut on 2×6 plates	R-15 batt woven between the faces
Plates	2×6: 1,109 lf · 2×4: 998 lf	one bottom, two top	buy 16' lengths: 77 × 2×6, 69 × 2×4 (10% waste)
Wall sheathing	100 sheets 7/16" OSB 4×8	exterior walls incl. garage, rim to top plate, 10% waste	plus housewrap
Blocking	80 lf of bands	19 bands (blocking schedule)	2×6 to 2×10 offcuts
Roof trusses	67 trusses	roof framing schedule below	priced by the truss maker from their layout
Patio and porch rafters	45 rafters	2×10 patio, 2×8 porch, at 24" o.c.	plus beams, ledger and hangers
Drywall, walls	7,580 sf · 174 sheets 1/2" 4×12	house walls, both faces of partitions, less openings	
Drywall, ceilings	3,010 sf · 69 sheets 5/8" ceiling board 4×12	flat 2,067 sf + vault 943 sf (4:12)	trusses at 24" o.c.: 5/8" board
Drywall, garage	1,272 sf · 30 sheets 5/8" Type X 4×12	walls to the house and the ceiling	tape and seal for the house separation
Tile backer	272 sf cement board	two showers to 8', tub surround to 6'	pedestal tub needs none
Studs, total	1,002 pcs	kings, jacks and corners included	add 10% for blocking and waste